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MARCIA™ WHITE PAPER

What Video Surveillance Can Learn From Hollywood About Storing Video

Hollywood has trusted LTO with its archives for decades. Why has surveillance been left behind, and what changes now?



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Two industries, one problem

Dailies. B-roll. Circle takes. For more than a decade, film and television production has lived with an explosion of recorded video. In the days of film, whole sections of a movie ended up on the cutting-room floor. Once digital cinema cameras removed the cost of film stock, productions began to keep everything: every take, every angle, every rehearsal. Many in Media & Entertainment (M&E) point to the arrival of digital cameras such as RED, first conceived in 2005, as the tipping point. One side effect followed: a lot of extra video.

Video surveillance is M&E's sister industry. It, too, records enormous amounts of video every day, most of which nobody watches until something happens. And it, too, must keep that video for months or years: for investigations, litigation, compliance and, increasingly, for new analytics that can learn from history.

The difference is what each industry did about it. Hollywood built workflows to store terabytes and petabytes of video at the lowest possible cost and still get any clip back when an editor needs it. Surveillance mostly kept buying racks of spinning hard drives, then cut resolution, frame rate and retention when the budget ran out.

This paper looks at what Hollywood learned, why those lessons have been slow to reach surveillance, and how Cozaint's patented MARCIA™ platform finally makes them practical for security teams.

What Hollywood figured out

1. Keep the original, unaltered video

Studios do not throw footage away to save storage. A shot that was not used today may be reused in a sequel, a director's cut or a documentary years later. The original is an asset.

Surveillance video is the same kind of asset. The clip that matters is often the one nobody knew mattered on the day it was recorded: the slip-and-fall claim filed eight months later, the fraud pattern found next year, the investigation that reaches back further than anyone planned. Reducing resolution, frame rate or retention to fit the storage budget throws away exactly the evidence the cameras were bought to capture.

2. Use more than one tier of storage

Just like surveillance video, production footage is rarely viewed after the first few days. Hollywood's answer is a lifecycle: recent, active footage on fast disk where editors work with it; everything else on LTO data tape, where it costs very little to keep and draws no power when the cartridge is at rest.

Lifecycle stage	Hollywood	Surveillance
Capture	Digital cinema cameras on set	IP cameras recording 24/7
Active work	Editors cut today's footage on fast disk	Operators review recent incidents on disk
Long-term archive	Every take kept on LTO	Months or years of video kept on LTO
Finding a clip later	Search by metadata, recall from the archive	Search by camera, date and time; analytics; recall from the archive
Proof the clip is genuine	Original camera files preserved	Original, unaltered recording supports chain of custody

3. Add information that makes video findable

Hollywood tags footage with metadata. "Airplane flyover of the Golden Gate Bridge at sunset" is a clip a production company can find in seconds and reuse rather than paying a crew to shoot it again. In surveillance, video analytics plays the same role: people, vehicles, objects and events make years of footage searchable. The more searchable history becomes, the more valuable it is to keep.

4. If you cannot get it back quickly, it is not worth keeping

This is the lesson that matters most. Hollywood's archive works because an editor can bring back any clip without a research project. An archive nobody can use is just a cost.

Why LTO, and why it is not "your grandfather's tape"

When some people in security hear the word "tape," they think of VHS. LTO (Linear Tape-Open) has nothing in common with it. LTO is a digital data storage standard, first introduced in 2000 and developed by Hewlett Packard Enterprise, IBM and Quantum through the LTO Program. It is the de facto standard for data tape and is heavily used in Hollywood, broadcasting, research and enterprise IT.

What LTO offers	Why it matters for surveillance video
Very low cost per terabyte	Far less than keeping years of video on spinning disk or in the cloud
Large capacity per cartridge	LTO-10 stores 30 TB per cartridge, and up to 40 TB with the newest LTO-10 cartridges. The LTO roadmap extends to LTO-14 at a planned 576 TB

What LTO offers	Why it matters for surveillance video
Decades of archive life	Media manufacturers rate LTO for up to 50 years of archival life
Zero power at rest	A cartridge in a library slot or on a shelf draws no power and needs no cooling
Sequential data, like video	Video is written as a continuous stream, exactly the way tape records
Offline (air-gapped) copies	Cartridges can be removed and stored offline, out of reach of the network
Write-once (WORM) media and encryption	Options to keep recordings from being altered and to protect them from unauthorized access
Open format with LTFS	The Linear Tape File System makes a cartridge self-describing and readable on standard systems

Studies cited by the tape industry show that over time tape systems use about 87% less energy than an equivalent capacity of hard disk, and produce far less CO2e over their lifecycle. Many feature productions have also adopted daily archiving of on-set footage to LTO, because insurers and bonding companies trust the same tape platform that banks use.

Why surveillance was left behind

If LTO is so well proven, why is it not in every surveillance server room? Because earlier attempts to use tape in surveillance asked the wrong person to do the work.

The old ways of using tape

- **Pre-fetch.** Restore video from tape before you need it. That assumes the operator knows what they are looking for. In surveillance they usually do not; that is why they are reviewing video in the first place.
- **Backup and restore.** Copy video to tape like any IT backup, then restore it on request. The operator has to leave the video management software (VMS), file a request and wait for IT. Where does the restored video go, and which system plays it?
- **Multiple steps, IT involved.** Every extra step makes the archive less likely to be used.

The editor's timeline and the operator's timeline

A Hollywood editor and a surveillance operator are close cousins. Both work on a timeline. But there are two important differences:

	Editor's timeline (non-linear editor)	Operator's timeline (VMS)
Length	Finite: one production	Endless: every camera, every day
Does the user know what they need?	Yes, the editor knows the shot	Often not; the operator is searching
Who manages the archive?	Post-production staff	Should be nobody; the operator must stay in control

Security is not an IT process. In surveillance, the operator is king. For tape to work, the archive has to come to the operator's timeline, not the other way around.

What changes now: MARCIA

Cozaint MARCIA™ Intelligent Video Lifecycle Management removes the barrier that kept LTO out of surveillance. Its patented architecture (U.S. Patent No. 12,755,902) lets an operator's normal VMS playback request recall archived video from LTO and play it on the same timeline they use every day.

Request.
The operator moves the VMS timeline to a past date and time.

Locate.
MARCIA identifies which storage tier holds that video.

Map.
It maps the request to the exact cartridge and position.

Recall.
The tape library loads the cartridge and reads the video.

Play.
The video returns to the same VMS timeline.

Recent video plays from disk at once. Older video on LTO typically returns within a few minutes, with no extra steps, no restore job and no IT ticket. New video is replicated to LTO automatically in the background, so organizations keep months or years of video without adding racks.

Just like a Hollywood editor pulling the original camera file, the operator works with the original, unaltered recording, which supports chain of custody when video is submitted as evidence.

> **Competitors can sell tape libraries, archive software or disk.** What makes MARCIA different is the patented VMS integration that turns LTO into part of the operator's normal workflow. MARCIA works with Milestone XProtect, Hanwha WAVE, Digital Watchdog DW Spectrum, Network Optix Nx Witness and Cozaint BOBBYvms, and is validated on HPE StoreEver, Quantum Scalar and ATTO XstreamCORE.

The economics, in brief

The cost of a disk-only archive grows with every camera, every increase in resolution and every extra month of retention. It also keeps costing money after purchase: power and cooling 24/7, rack space, support and disk replacement every few years.

With a lifecycle approach, only the most recent video lives on disk. Everything else moves to LTO, where adding retention mostly means adding cartridges.

> "It is always difficult to find a system that fits within our budget and meets our long-term retention requirements. Cozaint's askALICE has been an ideal fit for our surveillance recording needs and our budget."
>
> Scott Prather, Senior IT Analyst, City of Oceanside, California

At the City of Oceanside's wastewater treatment plant, more than 30 cameras recording 1080p video 24/7 needed one year of retention, about 548 TB. A two-tier disk-and-LTO design built on the architecture MARCIA is based on came in at **54% lower five-year cost** than disk-only, as calculated with the FUJIFILM video surveillance retention cost calculator. The full story is in the City of Oceanside case study.

Lessons to take back to your organization

1. **Treat recorded video as an asset**, not a storage problem to be shrunk.
2. **Record at the quality your cameras were bought for**, then change the storage, not the evidence.
3. **Separate recent video from long-term retention.** Keep a right-sized disk tier and put the archive on LTO.
4. **Keep the operator in the VMS.** Any archive that needs IT or a separate tool will not be used.
5. **Compare five-year cost**, including power, cooling and disk replacement, not just the purchase price.
6. **Plan for the value of history.** Investigations, litigation, compliance and AI re-analysis all depend on video you kept.

Hollywood solved this problem years ago. MARCIA makes the same approach practical for surveillance.

See your own numbers. Run the storage cost calculator at **cozaint.com/assessment**, or talk to the Cozaint team at **info@cozaint.com** or **760-975-8000**.

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